Question:
Carlos is planting a lemon tree. The tree will cost $90 to plant. Each year it will grow 7 lemons, which he can sell for $1.5 each. It costs $3 a year to water and feed the tree. How many years will it take before he starts earning money on the lemon tree?

Reasoning + Answer:

<think> To determine how many years it will take before Carlos starts earning money on the lemon tree, we need to calculate the total yearly cost and compare it with the revenue from selling lemons. The initial cost to plant the lemon tree is $90. The yearly cost to water and feed the tree is $3. Each year, the lemon tree produces 7 lemons, and each lemon is sold for $1.5.

Let's denote the number of years as n. The total yearly cost includes the initial planting cost and the yearly maintenance cost. The revenue from selling lemons each year is 7×1.5. We need to find the smallest n for which the revenue exceeds the total yearly cost.

The total yearly cost is:
$$90 + 3n$$

The revenue from selling lemons each year is:
$$7 \times 1.5 = 10.5$$

We need to find n such that:
$$10.5 \geq 90 + 3n$$

Rearranging the inequality to solve for n:
$$10.5 - 90 \geq 3n$$
$$-79.5 \geq 3n$$
$$n \geq -\frac{79.5}{3}$$
$$n \geq -26.5$$

Since n must be a positive integer, we need to find the smallest positive integer n that satisfies this inequality. Clearly, n must be at least 10 because:
$$10 \times 3 = 30$$
$$10.5 > 90 + 30$$
$$10.5 > 120$$

Let's check the first year:
$$\text{Year 1 Revenue} = 7 \times 1.5 = 10.5$$
$$\text{Year 1 Cost} = 90 + 3 = 93$$
$$10.5 < 93$$

Let's check the second year:
$$\text{Year 2 Revenue} = 7 \times 1.5 = 10.5$$
$$\text{Year 2 Cost} = 90 + 3 \times 2 = 96$$
$$10.5 < 96$$

Let's check the third year:
$$\text{Year 3 Revenue} = 7 \times 1.5 = 10.5$$
$$\text{Year 3 Cost} = 90 + 3 \times 3 = 102$$
$$10.5 < 102$$

Let's check the fourth year:
$$\text{Year 4 Revenue} = 7 \times 1.5 = 10.5$$
$$\text{Year 4 Cost} = 90 + 3 \times 4 = 102$$
$$10.5 < 102$$

Let's check the fifth year:
$$\text{Year 5 Revenue} = 7 \times 1.5 = 10.5$$
$$\text{Year 5 Cost} = 90 + 3 \times 5 = 115$$
$$10.5 < 115$$

Let's check the sixth year:
$$\text{Year 6 Revenue} = 7 \times 1.5 = 10.5$$
$$\text{Year 6 Cost} = 90 + 3 \times 6 = 120$$
$$10.5 < 120$$

Let's check the